DAY 24 · THEORY HANDOUT

Turn Replies Into Checked Data

Compare plain text, JSON, Pydantic schemas, and list parsing.

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The idea and its useful terms

Compare plain text, JSON, Pydantic schemas, and list parsing.

JSON

Text with named fields that programs can read.

Schema

A rule for field names and allowed value types.

Validation

Check whether a value fits the declared rules.

A program needs fields it can use, not a long paragraph it must guess how to read. JSON gives us named values.

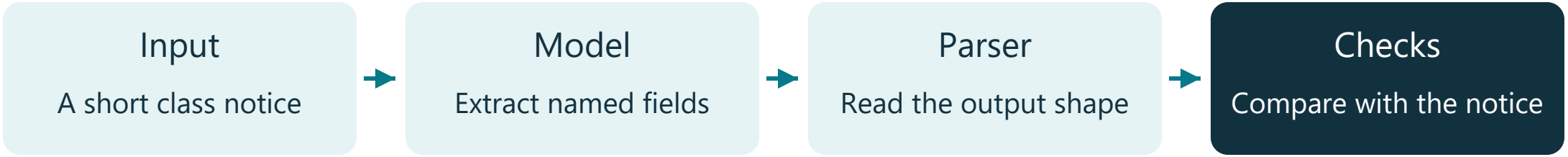
A valid schema proves the shape and types. It does not prove that the model copied the correct facts.

> Structured output reduces format mistakes. A wrong fact can still fit a perfect schema.

Follow the work, one step at a time

Each step has a job. We can inspect its output before moving on.

Read the path from left to right.



Each arrow means: pass the result to the next step.

A parser reads an output representation. JSON syntax uses quoted keys, values, and braces. A schema then specifies which fields and value types are expected. Parsing and validation are separate checks.

For example, room can be a valid integer but the wrong room. Compare the field with the source notice after validation. Some schemas convert compatible types, so a successful parse is not a complete fact check.

> If the answer is wrong, inspect the input and the intermediate results. The last sentence alone cannot show which step failed.

A small example we can check

The values here are classroom examples. The notebook shows actual results.

Source

Python, Monday, 10:00.
Room 2.

JSON fields

course: Python
room: 2

Two checks

Is room an integer?
Does source say room 2?

Read the given input first. Point to the fact or value needed for the requested output.

Run the matching example, then try the fresh input at the end of the notebook. Compare both results with the known input.

> Structured output reduces format mistakes. A wrong fact can still fit a perfect schema.

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Choose the setting or tool for its job

A different choice changes the work. These are the choices we compare.

Text parser

Use for a written reply.

No field validation.

JSON parser

Use for flexible fields.

Check missing keys.

Pydantic schema

Use for fixed typed fields.

Reject invalid values.

Parse a small known JSON string before asking the model to produce one.

Use a schema with room as an integer. Test a wrong type.

> Extract a fresh notice and compare every field with the source. Missing facts should stay missing.

Practice, check, and read the full lesson

Use the notebook to try the steps. Use the article when you need the longer explanation.

1. Build

Run the small example.

Inspect each step.

2. Change

Try the fresh input.

Keep the goal clear.

3. Check

Compare with the source.

Keep mistakes visible.

Extract a fresh notice and compare every field with the source. Missing facts should stay missing.

Read the complete tutorial breakdown for this section in our KGP Talkie article.

> Structured output reduces format mistakes. A wrong fact can still fit a perfect schema.

DAY 24 · COMPLETE

Thank You

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